Machine Learning Generalization

A short source for PDF parsing and Knowledge Map tests.

1. Generalization and Data Splits

Generalization is a model's ability to perform well on examples that were not used during training. Training error measures fit on observed data, while validation error estimates behavior on held-out data.

A training set is used to fit parameters. A validation set helps compare choices such as model complexity and regularization. A test set should remain untouched until the final evaluation.

If decisions repeatedly use the test set, the test result becomes optimistic. The separation between training, validation, and test data protects the credibility of the final estimate.

2. Overfitting and Its Evidence

Overfitting occurs when a model captures details or noise that do not generalize. The model may achieve very low training error but substantially higher validation error.

A widening training-validation gap is evidence of overfitting. Learning curves can reveal this gap as training continues or as model capacity increases.

High training accuracy alone does not prove that a model is good. Performance must be checked on unseen data, and data leakage must be ruled out before interpreting the result.

3. Regularization and Early Stopping

Regularization discourages unnecessarily complex solutions. L1 and L2 penalties add a cost for large parameter values, while dropout reduces reliance on individual activations.

Early stopping monitors validation performance and stops training when further optimization begins to reduce generalization. It is a practical control for iterative learning algorithms.

A mitigation should match the evidence. Teams can simplify the model, collect more representative data, add regularization, or improve the validation procedure. No single technique guarantees better generalization in every setting.